

Jinning LIANG

Address: Kavli Institute for Astronomy and Astrophysics 404, Peking University, P.R.C.

Contact: (+86) 18977285216 jinningliangastro@gmail.com

ORCID ID: 0000-0001-8405-2921 Personal Website: jinningliang.com

RESEARCH INTERESTS

Keywords: galaxy formation theory, galaxy-halo connection, AGN feedback, galactic dynamics, galactic chemical evolution, cosmology, numerical simulation, semi-analytical model

EDUCATION

The Kavli Institute for Astronomy and Astrophysics, Peking University <i>Ph.D. student in Astrophysics, Advisor: Prof. Fangzhou Jiang</i>	Sep 2025 - Sep 2029
The Institute for Computational Cosmology, Durham University <i>STFC-funded Ph.D. student in Physics, Advisor: Prof. Cedric Lacey</i>	Oct 2023 - Mar 2025
<i>M.Sc. in Physics received</i>	Oct 2023 - Sep 2024
School of Physics and Technology, Wuhan University <i>Bachelor of Science in Physics</i>	Mar 2025
➤ Astronomy Class (Selected from pool of 280 students due to outstanding performance and enthusiasm for astronomy)	Sep 2019 - Jun 2023
➤ GPA: 3.80/4.00; 90.15/100; Ranking: 1/196	Sep 2020 - Jun 2023

PUBLICATIONS (* Denotes co-first author, † Denotes corresponding author, Total Citation: 57, h-index: 6)

- [1] Jinning Liang, C. G. Lacey[†] et al., *Simulating AGN wind feedback with variable feedback efficiencies in idealised disc galaxies*, Submitted to MNRAS, 2026 [[2603.12322](#)]
- [2] Jinning Liang, F. Jiang[†], H.J. Mo, et al., *Connection between galaxy morphology and dark-matter halo structure II: predicting disk structure from dark-matter halo properties*, ApJ **1003** (2026) 12, 2026 [[2512.13822](#)]
- [3] F. Jiang^{†,*}, Jinning Liang^{*} et al., *Formation and Environmental Context of Giant Bulgeless Disk Galaxies in the Early Universe: Insights from Cosmological Simulations*, Submitted, 2025 [[2504.01070](#)]
- [4] Zou, S[†], Simcoe, R.A., Petitjean, P., Péroux, C., Champagne, J., Wang, F., Jinning Liang et al., *Disturbed cold gas in galaxy and structure formation*, ApJ **994** (2025) 53 [[2502.14705](#)]
- [5] Jinning Liang, F. Jiang[†], H.J. Mo et al., *Connection between galaxy morphology and dark-matter halo structure I: a running threshold for thin discs and size predictors from the dark sector*, MNRAS **541** (2025) 2304 [[2403.14749](#)]
- [6] Jinning Liang, F. Jiang[†] et al., *Constrain the Dark Matter Distribution of Ultra-diffuse Galaxies with Globular-Cluster Mass Segregation: A Case Study with NGC5846-UDG1*, ApJ **964** (2024) 53 [[2304.14431](#)]
- [7] Jinning Liang, E. Gjergo[†] & X. Fan, *Assessing stellar yields in Galaxy chemical evolution: benchmark on observational stellar abundance patterns*, MNRAS **522** (2023) 863 [[2304.00208](#)]
- [8] E. Gjergo[†], A.G. Sorokin, A. Ruth, E. Spitoni, F. Matteucci, X. Fan, Jinning Liang et al., *GalCEM I - A Publicly-Available Detailed Isotopic Chemical Evolution Code*, ApJS **264** (2023) 44 [[2301.02257](#)]
- [9] H. Liu^{*}, Jinning Liang^{*} & J. Jia[†], *Deflection and Gravitational lensing of null and timelike signals in the Kiselev black hole spacetime in the weak field limit*, Class. Quantum. Grav **39** (2022) 195013 [[2204.04519](#)]

TEACHING

PHYS2631 Stars & Galaxies, Durham University <i>Workshop Demonstrator</i>	Oct 2023 - Apr 2024
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SCIENTIFIC OUTREACH

Celebrate Science, Durham University <i>Educated children with basic knowledge and illustration about gravitational lensing and galaxy formation</i>	2023
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TALKS (* Denotes invited talk)

[1] Workshop Talk, University of California Santa Cruz, Santa Cruz, United States <i>Disk formation in halo conditions</i>	2026
[2] Seminar Talk, The University of Western Australia, Perth, Australia <i>Disk formation in halo conditions</i>	2026
[3] Conference Talk, The 3 rd Shanghai Assembly on Cosmology and Structure Formation, Shanghai, China <i>Connection between galaxy morphology and dark-matter-halo structure</i>	2025
[4] Conference Talk, CSST annual meeting, Kunming, China <i>Galaxy-halo coevolution: simulations and mock observations – Halo condition on disc formation</i>	2025
[5] Seminar Talk*, The Chinese University at Hongkong, Hongkong, China <i>Connection between galaxy morphology and dark matter halo structure – Halo condition on disc formation</i>	2025
[6] Seminar Talk, Flat Talk, Durham, UK	2024

AGN feedback by line-driven winds

[7] Workshop Talk, 20th Durham-Edinburgh eXtragalactic (DEX) Workshop, Durham, UK 2024

New morphological decomposition method and varying circularity threshold

[8] **Seminar Talk***, Shanghai Astronomical Observatory, Shanghai, China 2023

Connection between galaxy morphology and dark-matter halo structure

[9] **Seminar Talk***, Shanghai Jiaotong University, Shanghai, China 2023

Dark matter properties and its connection with galaxy morphology

[10] Seminar Talk (online), University of Arizona, Arizona, AZ, USA 2022

Globular Clusters in UDGs

[11] Conference Talk, ISM Physics and Chemistry Conference, Yichang, Hubei, China 2022

Galactic stellar abundance scatter investigated through yield analysis in galaxy chemical evolution

SELECTED HONORS AND AWARDS

Science and Technology Facilities Council Scholarship 2023

Yu Gang - Song Xiao Scholarship of Wuhan University 45/30000 2022

First-class Scholarship of Wuhan University Top 5% 2022

MCM&ICM Finalist Award Top 2% 2022

National Astronomical Observatories Scholarship 3/600 2021

SKILLS

Programming Languages & Software: Python (Extensively), Mathematica (Extensively), Matlab and LaTeX

Language: Mandarin (Native), English (Proficient)

Simulations & Semi-analytical model: *IllustrisTNG*, *EAGLE*, *Swift-COLIBRE*, *NuPyCEE* and *SatGen*

MENTORSHIP

[1] **Haiyang Xing**, postgraduate at Peking University since Jan 2024

Finding Milky Way catalog using galactic decomposition method (MorphDecom), in preparation

[2] **Peng Xu**, undergraduate at Tsinghua University since Mar 2024

Classifying and visualizing cosmic web with T-web method and DisPerSE

[3] **Yalin Wu**, undergraduate at Peking University since Oct 2024

Simulating stellar streams and globular clusters in semi-analytical model, in preparation

[4] **Yiheng Tian**, undergraduate at Peking University Dec 2024 - May 2025

Studying the evolution of bars quantities in IllustrisTNG, undergraduate thesis

[5] **Jianyuan Luo**, postgraduate at Peking University since Mar 2025

Finding inner mini-disk of massive galaxies at high redshift using cosmological simulations, in preparation

[6] **Di Yuan Wang**, Internship student at Peking University since Apr 2026

Connection between galaxy spheroids and halo conditions, in preparation

VISITING EXPERIENCES

[1] Internation Centre for Radio Astronomy Research, The University of Western Australia Jul 2026

[2] The Kavli Institute for Astronomy and Astrophysics, Peking University Mar 2023 - Oct 2024

[3] Shanghai Astronomical Observatory Jan 2022 - Feb 2022

PROFESSIONAL REFERENCES

[1] **Prof. Fangzhou Jiang**

The Kavli Institute for Astronomy and Astrophysics, Peking University

Email: fangzhou.jiang@pku.edu.cn

[2] **Prof. Cedric Lacey**

The Institute for Computational Cosmology, Durham University

Email: cedric.lacey@durham.ac.uk

[3] **Dr. Sownak Bose**

The Institute for Computational Cosmology, Durham University

Email: sownak.bose@durham.ac.uk

[4] **Prof. Houjun Mo**

Department of Astronomy, University of Massachusetts

Email: hjmo@umass.edu

[5] **Prof. Xilong Fan**

School of Physics and Technology, Wuhan University

Email: xilong.fan@whu.edu.cn